

# Executive Summary: Dropout Prediction Analysis

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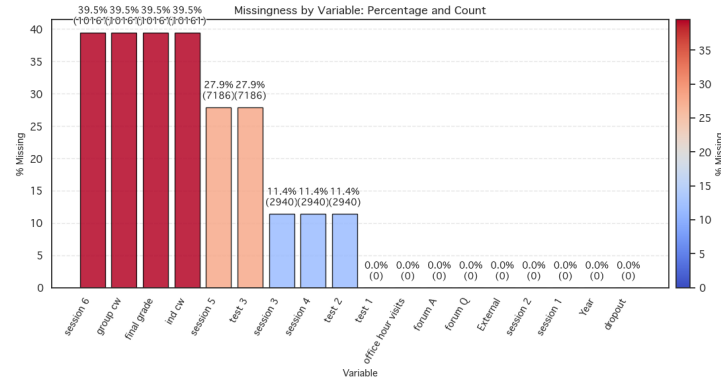
## 1 Introduction

Predicting student dropout is of paramount importance to educators and institutions alike, as attrition carries substantial academic, social, and financial consequences [1]. Early identification of at-risk students enables the timely deployment of targeted support interventions, including academic advising, mentoring, and financial assistance [2].

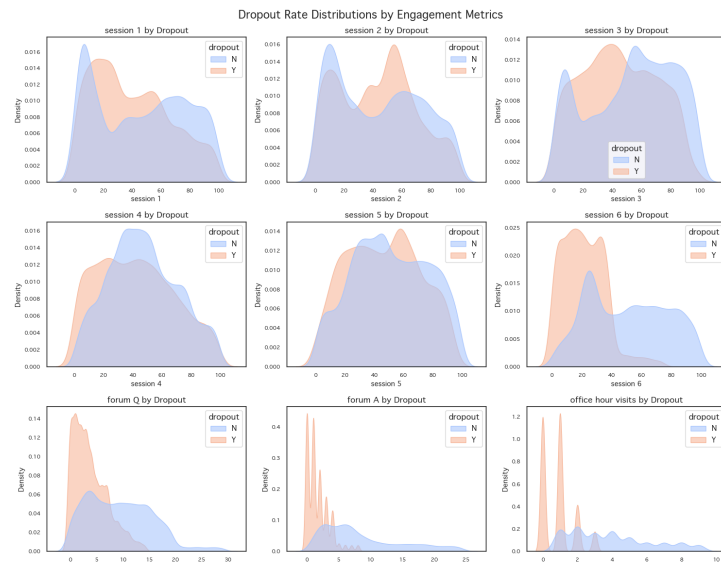
## 2 Methodology

### 2.1 Exploratory Data Analysis

Missing data patterns reflect programme progression rather than random omission, with later-stage variables showing higher missingness as students disengage [1]. Missingness therefore carries predictive information about dropout: it should be treated as a meaningful signal, not a data quality issue. Engagement patterns between persisting and dropout students diverge progressively over time [3], with later stages showing clear separation as dropout students cluster at low engagement levels [4, 5]. Voluntary metrics such as forum participation and office hour attendance prove especially discriminating, suggesting active participation is a more sensitive early-warning indicator of attrition than passive attendance alone.

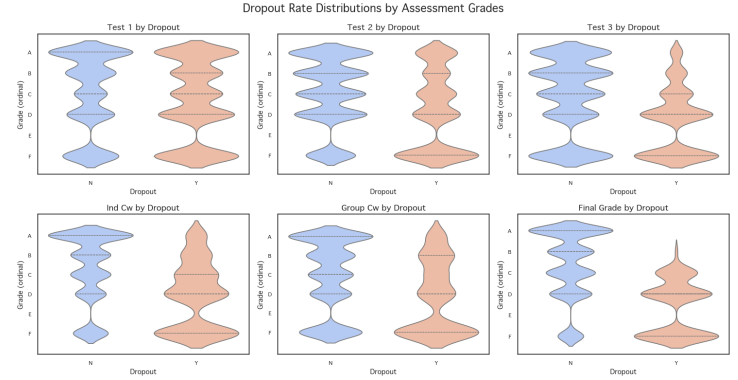


**Figure 1:** Missing rates by variable. Detailed breakdown provided in the notebook.



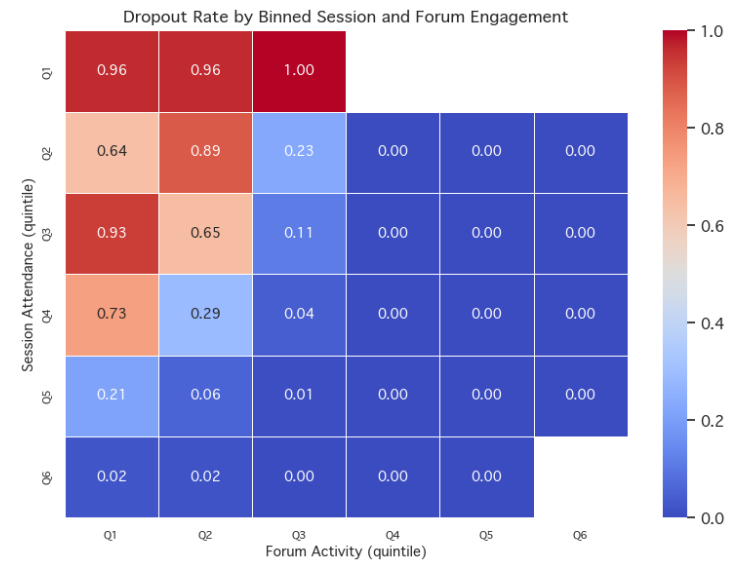
**Figure 2:** Engagement distributions by dropout status. Extended interpretation is provided in the accompanying notebook.

dropout students, with early assessments showing similar outcomes before later stages reveal a clear concentration of low grades among eventual dropouts [6]. By the final stage, dropout students are almost entirely associated with failing outcomes.



**Figure 3:** Assessment-grade distributions stratified by dropout outcome, showing progressively stronger concentration of low and failing marks among eventual dropouts in later evaluations. From a business and retention-management perspective, this pattern supports using mid-term assessment deterioration as an operational trigger for targeted interventions (for example, proactive advising, academic support, and outreach), where earlier action can reduce downstream attrition costs and protect tuition-revenue continuity.

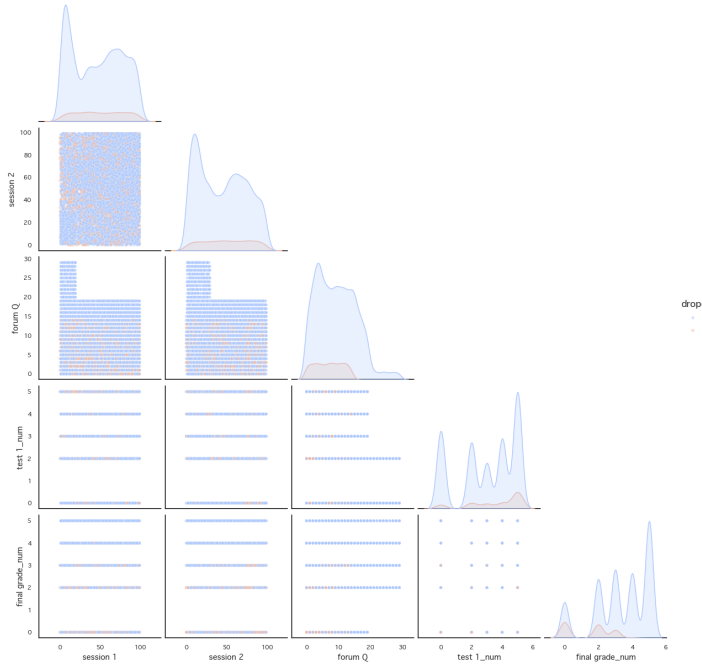
Heatmap analysis reveals a strong interaction between attendance and forum activity, with a clear high-risk zone emerging among students low in both metrics [3]. As either measure increases, attrition declines sharply, and higher forum participation appears to mitigate risk even at moderate attendance levels, suggesting voluntary engagement acts as a protective factor and that combined behavioural signals are more informative than either metric alone.



**Figure 4:** Stage-binned dropout landscape by session attendance and forum engagement quintiles. The heatmap reports the conditional dropout rate for each joint engagement bin, showing a pronounced high-risk zone (dark red) at low attendance and low forum activity, and a sharp risk decline (deep blue) as either dimension increases, consistent with voluntary engagement acting as a protective factor.

Academic performance diverges progressively between persisting and

Pairwise Relationships: Key Features vs. Dropout



**Figure 5:** The pairwise plots of key features, by dropout. Participation patterns between persisting and dropout students shift from overlapping in early sessions to highly distinct distributions by Session 6. Active behaviours, such as forum engagement and office hour attendance, provide the strongest signal for identifying at-risk students, as dropout populations exhibit almost no activity in these voluntary categories compared to their persisting peers.

## 2.2 Segmentation Analysis

K-means clustering on PCA-reduced features (see Figure 6) revealed four distinct student profiles summarised in Table 1. Clusters 0 and 3 identify “High-Risk” segments, characterised by minimal forum activity and alarming dropout rates of 93.5% and 78.0%, respectively (visualised in Figure 7). Conversely, Clusters 1 and 2 represent “Highly Engaged” students; Cluster 2, notably, shows the highest forum interaction and near-zero attrition (0.8%).

**Table 1:** Mean Student Metrics by Cluster

| Clust. | Count | Sess.  | Forum | Assess. | Engage. | Drop % |
|--------|-------|--------|-------|---------|---------|--------|
| 0      | 6,526 | 137.86 | 4.37  | 0.92    | 153.04  | 93.5%  |
| 3      | 5,886 | 174.50 | 5.61  | 1.31    | 229.56  | 78.0%  |
| 1      | 8,936 | 319.28 | 16.80 | 3.08    | 984.99  | 13.7%  |
| 2      | 4,401 | 295.03 | 23.13 | 2.97    | 880.72  | 0.8%   |

Header definitions: *Sess.*: Session Total; *Forum*: Forum Total; *Assess.*: Assessment Mean; *Engage.*: Engagement Grade; *Drop %*: Dropout Rate.

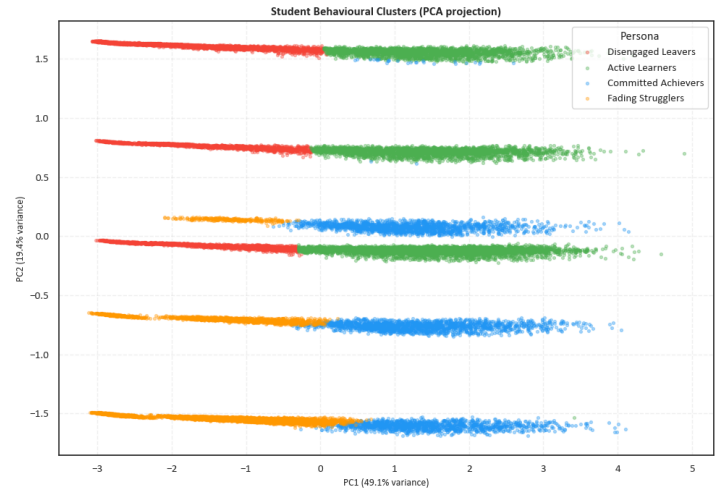
## 2.3 Data Pre-processing

Data was standardised, encoded, and stratified for training [7]; see notebook for full pre-processing details.

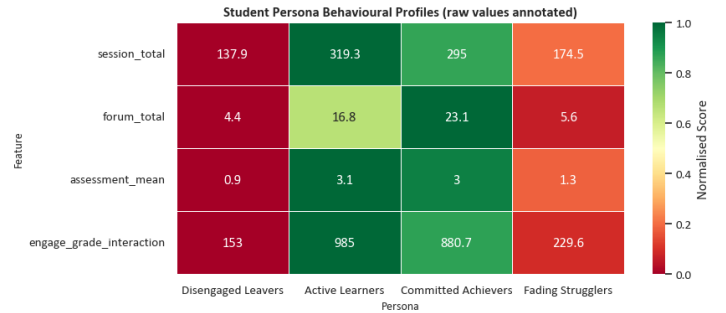
## 2.4 Survival Analysis

### 2.4.1 Motivation

Unlike standard classification, survival analysis explicitly models the timing and stages of dropout, which is important given that missingness in our dataset is non-random: students who leave cease generating new data, producing left-censored records. Aligning the analysis with tuition-fee checkpoints and academic milestones allows identification of when students are most at risk, enabling stage-specific interventions.



**Figure 6:** Principal Component Analysis (PCA) of student engagement and performance intensity. The **horizontal axis (PC1)** defines a primary gradient between high-achieving, active students and those exhibiting low engagement and weaker academic outcomes; the **vertical axis (PC2)** distinguishes between differing behavioural styles, such as structured session attendance versus self-directed forum participation. This mapping reveals that dropout-prone personas occupy coherent, non-random regions in behavioural space, facilitating a segmented retention strategy. By monitoring a student’s movement across these dimensions, the business can prioritise resource allocation toward high-risk segments and deploy targeted interventions: ranging from automated nudges to mandatory tutor check-ins—before attrition occurs.



**Figure 7:** Heatmap of student persona behavioural profiles across four distinct segments, illustrating the relationship between engagement metrics and academic outcomes. The visualisation reveals a clear performance gradient, where ‘Committed Achievers’ and ‘Active Learners’ demonstrate high session attendance and forum activity, reflected in superior assessment means. Conversely, ‘Disengaged Leavers’ exhibit critical deficits across all features, while ‘Fading Strugglers’ present a high-risk profile of moderate attendance coupled with low attainment. By identifying these specific behavioural signatures, the business can customise intervention strategies, moving from broad retention policies to targeted support tailored to the unique participation patterns and academic needs of each student persona.

### 2.4.2 Staged Cohort Construction

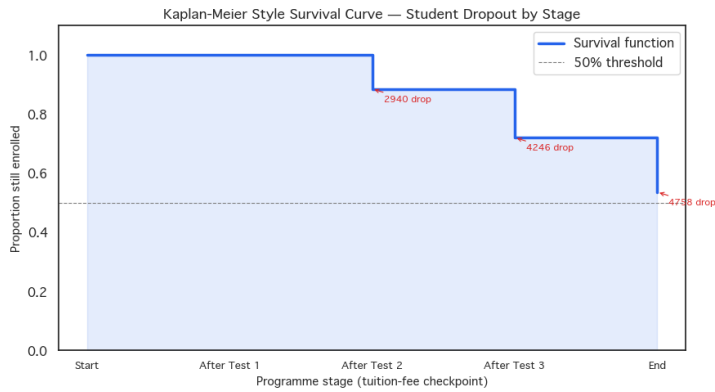
Four longitudinal stages were defined, each corresponding to a set of available features and a tuition-fee checkpoint:

**Table 2:** Programme stages and available features at each checkpoint.

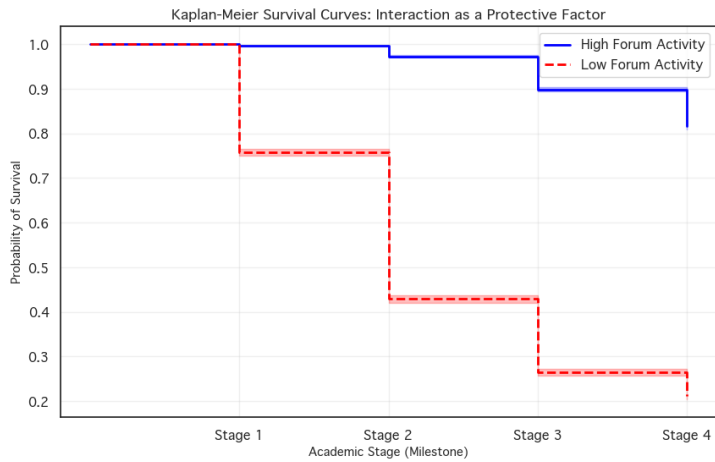
| Stage   | Columns Available                                        | Cohort       |
|---------|----------------------------------------------------------|--------------|
| Stage 1 | Session 1, 2; Test 1; Forum/Office-Hours (amortised ÷ 4) | Full cohort  |
| Stage 2 | + Session 3, 4; Test 2; Forum/Office-Hours (÷ 2)         | Pass Stage 1 |
| Stage 3 | + Session 5; Test 3; Forum/Office-Hours (× 3/4)          | Pass Stage 2 |
| Stage 4 | + Ind CW, Group CW, Final Grade; Full Forum/Office-Hours | Pass Stage 3 |

### 2.4.3 Kaplan–Meier Survival Curve

A Kaplan–Meier style survival curve tracks the fraction of the cohort still enrolled after each milestone, mirroring the approach used in clinical trials.



**Figure 8:** Kaplan–Meier survival function for student persistence over programme progression. The curve shows the estimated probability of remaining enrolled (not dropping out) as students advance through successive checkpoints; downward steps indicate dropout events, and steeper declines correspond to higher attrition intensity during that interval.

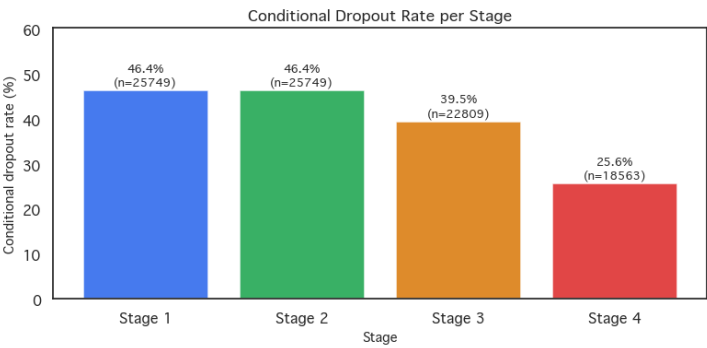


**Figure 9:** Stratified survival curves comparing persistence trajectories across student groups. Differences in curve separation indicate heterogeneous dropout risk over time, where groups with earlier and sharper declines exhibit lower retention and higher conditional attrition at stages.

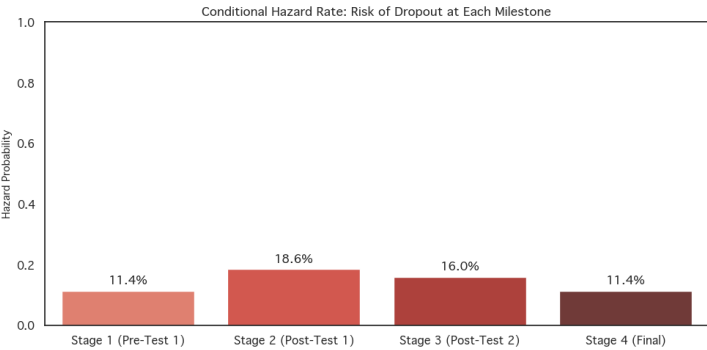
#### 2.4.4 Staged Predictive Modeling

**Table 3:** Sequential Predictive Modelling Architecture by Stage. The predictive framework employs an iterative approach where a dedicated model is trained at each milestone using only the temporal data available to that point. By training on the specific sub-cohort that has persisted to each stage, the system accounts for survival bias and progressively increases model complexity, moving from interpretable linear baselines to sophisticated ensembles, to capture the evolving behavioural signals of at-risk students.

| Stage   | Model Architecture          | Key Rationale & Benefit                                                                                                                                                                         |
|---------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stage 1 | Logistic Regression         | High interpretability; suitable for low-complexity early data.                                                                                                                                  |
| Stage 2 | Random Forest               | Captures non-linear relationships and feature interactions.                                                                                                                                     |
| Stage 3 | Histogram Gradient Boosting | Handles mid-programme complexity efficiently by leveraging native missing-value support and non-linear boosting, enabling robust detection of disengagement patterns without manual imputation. |
| Stage 4 | Stacking Ensemble           | Maximises predictive power by leveraging all cumulative signals.                                                                                                                                |



**Figure 10:** Conditional dropout rate by programme stage. Each bar represents the probability of dropout among students who remained enrolled up to that checkpoint, highlighting where attrition risk concentrates.

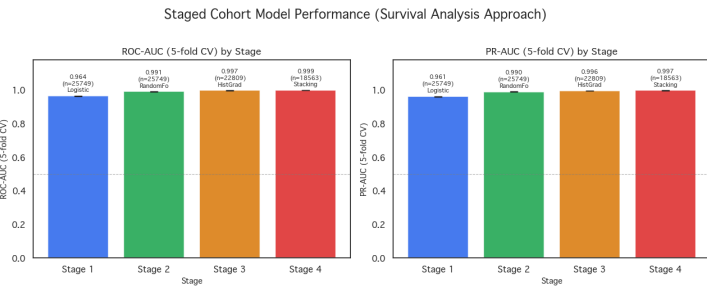


**Figure 11:** Estimated conditional hazard of dropout across stages. Higher bars indicate greater instantaneous risk of withdrawal at that stage, given prior persistence, and help identify the most critical intervention windows.

#### 2.4.5 Cross-Stage Performance

**Table 4:** Cross-stage model performance and conditional dropout dynamics across the programme timeline. ROC-AUC and PR-AUC summarise discriminatory power at each milestone, while the conditional dropout rate reports the proportion of students who withdraw among those still active at that stage. Together, these metrics show how predictive difficulty and attrition intensity evolve over time, supporting stage-specific intervention planning and resource allocation.

| Stage   | Model                | ROC-AUC (mean $\pm$ std) | PR-AUC (mean $\pm$ std) | Conditional Dropout Rate |
|---------|----------------------|--------------------------|-------------------------|--------------------------|
| Stage 1 | Logistic Regression  | 0.964 $\pm$ 0.002        | 0.961 $\pm$ 0.003       | 46.4%                    |
| Stage 2 | Random Forest        | 0.988 $\pm$ 0.001        | 0.982 $\pm$ 0.001       | 39.5%                    |
| Stage 3 | Hist. Grad. Boosting | 0.995 $\pm$ 0.000        | 0.987 $\pm$ 0.001       | 25.6%                    |
| Stage 4 | Stacking Ensemble    | 0.998 $\pm$ 0.001        | 0.983 $\pm$ 0.005       | 11.4%                    |



**Figure 12:** Barplots of ROC-AUC and PR-AUC for each stage.

Further details on interpretation in the notebook.

### 3 Results

#### 3.1 Evaluation Metrics

Models are evaluated primarily on dropout recall — minimising missed at-risk students — balanced against precision to avoid alert fatigue, with F1, ROC-AUC, and PR-AUC providing holistic performance benchmarks.

#### 3.2 Model Comparison

The optimised Random Forest uses a large ensemble ( $n\_estimators = 500$ ) with deep, controlled trees (see Table 5).

Table 5: Best Hyperparameters for the Random Forest Model

| Hyperparameter    | Value  |
|-------------------|--------|
| max_depth         | 20     |
| max_features      | 'sqrt' |
| min_samples_leaf  | 1      |
| min_samples_split | 5      |
| n_estimators      | 500    |

The tuned Random Forest achieved near-perfect, balanced performance across both classes (Table 6), with precision, recall, and F1 all around 0.99, indicating the model reliably identifies at-risk students without generating excessive false alarms.

Table 6: Classification Report for Random Forest Model

| Class        | Precision | Recall | F1-score | Support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.988     | 0.994  | 0.991    | 2761    |
| 1            | 0.993     | 0.986  | 0.989    | 2389    |
| Accuracy     |           |        | 0.990    | 5150    |
| Macro Avg    | 0.990     | 0.990  | 0.990    | 5150    |
| Weighted Avg | 0.990     | 0.990  | 0.990    | 5150    |

Both MLP variants achieve similar accuracy (Table 7), with deeper architecture offering only marginal gains, suggesting that additional complexity adds little once pre-processing and feature engineering are applied. This highlights the benefits of a data-centric approach.

Table 7: Performance Comparison of PyTorch MLP Architectures

| Model       | Class    | Precision | Recall | F1-score | Support |
|-------------|----------|-----------|--------|----------|---------|
| MLP Shallow | 0        | 0.982     | 0.985  | 0.984    | 2761    |
|             | 1        | 0.983     | 0.979  | 0.981    | 2389    |
|             | Accuracy |           | 0.983  |          | 5150    |
| MLP Deep    | 0        | 0.982     | 0.986  | 0.984    | 2761    |
|             | 1        | 0.984     | 0.979  | 0.982    | 2389    |
|             | Accuracy |           | 0.983  |          | 5150    |

### 4 Conclusion

This report demonstrates that dropout can be predicted with high accuracy by combining engagement signals, assessment outcomes, and staged modelling aligned to the course timeline, with the Random Forest delivering the strongest precision-recall balance and neural baselines providing confirmatory results. Declining participation, weaker academic performance, and reduced voluntary engagement emerge as the central risk drivers. Together, these findings support a practical early-warning framework enabling institutions to intervene earlier, allocate resources more effectively, and target support at the points of highest attrition risk.

#### 4.1 Limitations & Future Work

The primary limitations are single-institution scope restricting generalisability, a survival-stage table requiring empirical population before operational use, and the absence of non-academic predictors such as financial hardship and wellbeing indicators. Future work should prioritise external validation across cohorts, prospective intervention testing, and subgroup fairness checks, alongside time-to-event modelling and confidence interval reporting. Ultimately, linking model outputs to concrete intervention pathways — academic advising, attendance outreach, and tailored feedback — would translate predictive performance into measurable retention impact.

Appendices

Appendix A: Data Dictionary

Table 8: Data Dictionary of key variables.

| Category                | Variable Name     | Description                                                              | Values / Units               |
|-------------------------|-------------------|--------------------------------------------------------------------------|------------------------------|
| Student Characteristics | External          | Indicator of external vs. internal enrollment. Undergraduate study year. | 'Y' and 'N'                  |
|                         | Year              |                                                                          | Numeric                      |
| Engagement Metrics      | Session 1-6       | Percentage of each Zoom session viewed.                                  | 0-100% (Percentage)          |
|                         | Forum Q           | Number of question posts.                                                | Count                        |
|                         | Forum A           | Number of answer posts.                                                  | Count                        |
|                         | Office hour visit | Number of office hours attended.                                         | Count                        |
| Assessment Performance  | Test 1-3          | Test grades.                                                             | Grade/Score                  |
|                         | ind cw            | Individual coursework grade.                                             | Grade/Score                  |
|                         | group cw          | Group coursework grade.                                                  | Grade/Score                  |
|                         | Final grade       | Final exam grade.                                                        | Grade/Score                  |
| Target Variable         | Dropout           | Student programme completion status.                                     | 'Y' (must have paid) and 'N' |

Appendix B: Predictive Modelling Architecture

To identify students at risk of attrition, a heterogeneous ensemble of machine learning architectures was developed, ranging from traditional statistical classifiers to deep neural networks. This multi-model approach allows for a comparative analysis of feature-linear relationships versus complex non-linear behavioural patterns.

4.1.1 Deep Learning Architecture: MLP-Dropout

A Deep Multi-Layer Perceptron (MLP) was implemented using the PyTorch framework to capture high-order interactions within the student engagement data. The architecture is specifically designed to prevent overfitting on the behavioural signals through a combination of depth and stochastic regularisation.

- **Input Layer:** Accepts the  $N$ -dimensional feature vector, which includes ordinal grade mappings, session attendance, and forum activity metrics.
- **Hidden Layers:** The network comprises three fully-connected layers with a tapering neuron distribution ( $128 \rightarrow 64 \rightarrow 32$ ). Each layer utilises a Rectified Linear Unit (ReLU) activation function to introduce non-linearity.
- **Regularisation:** Dropout layers with a probability of  $p = 0.25$  are interleaved between hidden units. This architectural choice forces the model to learn redundant representations of student behaviour, ensuring that predictions are not overly reliant on any single performance indicator.
- **Output Layer:** A single neuron with a Sigmoid activation function produces a probability distribution  $\hat{y} \in [0,1]$ , representing the likelihood of dropout.

4.1.2 Ensemble Stacking and Meta-Learning

Recognising that different models excel at capturing varied data distributions, a *Stacking Ensemble* was constructed to synthesize the predictive strengths of disparate algorithms.

4.1.3 Base Learners

The first level of the stack consists of three distinct "expert" models:

1. **Random Forest:** Leverages an ensemble of decision trees to capture non-linear decision boundaries and assess feature importance.
2. **Deep MLP:** Captures complex, high-dimensional interactions as described in the previous section.
3. **Logistic Regression:** Serves as a baseline to capture global linear trends in the data.

4.1.4 Meta-Classifier

The final prediction is adjudicated by a *Logistic Regression meta-learner*. This model is trained on the cross-validated out-of-fold (OOF) probability predictions generated by the base learners. By learning the optimal weighting for each expert's output, the meta-classifier minimizes the collective residual error and provides a more generalised prediction than any single constituent model.

Appendix C: SHAP Analysis

4.1.5 Why SHAP?

SHAP values offer a principled, game-theoretic approach to model interpretability. They attribute the prediction for each student to individual features, quantifying both the direction and magnitude of each feature's impact. This allows us to:

- Identify the most influential predictors of dropout
- Understand heterogeneity in risk factors across students
- Communicate model decisions to stakeholders in an interpretable way

4.1.6 SHAP Analysis Workflow

We focus on the final-stage (Stage 4) model, which leverages the richest set of features. The workflow is as follows:

1. Fit the best-performing model (e.g., Random Forest or Stacking Ensemble) on the Stage 4 cohort.
2. Use the SHAP `TreeExplainer` to compute SHAP values for a representative sample of students.
3. Visualise global and local feature importances.

4.1.7 Global Feature Importance

The bar plot below summarises the average absolute SHAP value for each feature, indicating its overall contribution to dropout risk across the cohort.

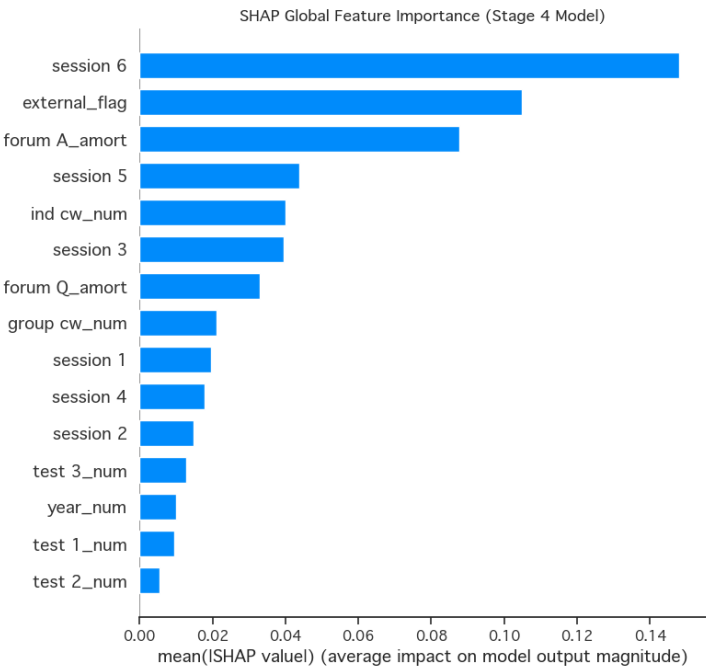
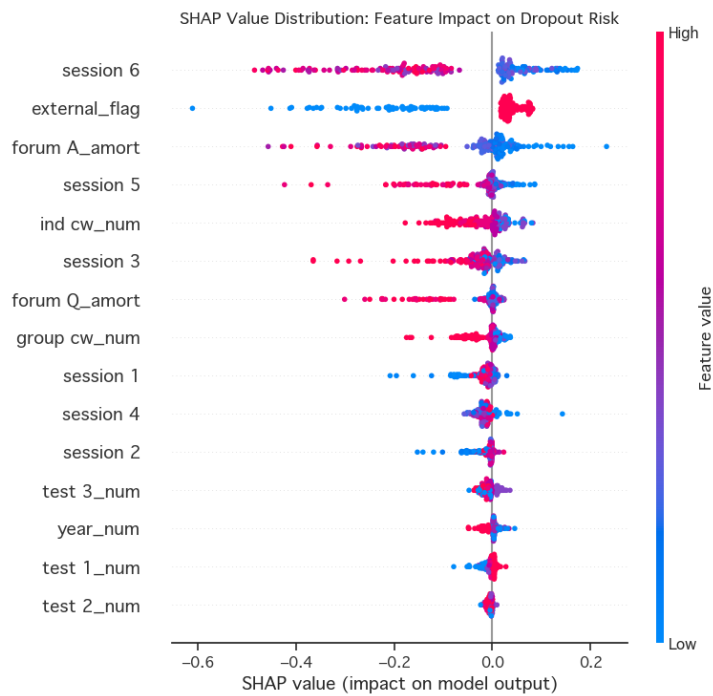


Figure 13: Global feature importance for dropout risk (mean absolute SHAP values).

4.1.8 Feature Impact Distribution

The SHAP summary plot below complements the global importance chart by showing how each feature behaves across individual students rather than only on average. This is particularly useful for distinguishing stable drivers of risk from variables whose influence depends on the student's broader behavioural profile. Features with a wide horizontal spread indicate substantial variation in contribution across the cohort, while tighter clusters suggest more consistent effects. The plot therefore helps identify whether dropout risk is driven by broadly shared patterns or by concentrated subgroups whose outcomes are shaped by specific combinations of engagement and performance signals.





**Figure 14:** SHAP value distribution for top features. Red indicates high feature values, blue indicates low. SFeatures are ranked on the y-axis by their total absolute impact (global importance). Each point represents an individual observation. The x-axis indicates the SHAP value: positive values (right of the center line) contribute to an increased risk of dropout, while negative values (left) contribute to a decreased risk. For example, high engagement in `session 6` and `forum A_amort` (red clusters on the left) is strongly associated with a lower risk of dropout. Conversely, high values for the `external_flag` (red cluster on the right) signify a significant increase in the predicted probability of dropout.

#### 4.1.9 Interpretation and Insights

- **Key drivers:** Features such as final grade, session attendance, and forum engagement consistently emerge as the most influential predictors of dropout risk.
- **Directionality:** Higher grades and engagement reduce dropout risk (negative SHAP values), while low engagement or poor grades increase it.
- **Heterogeneity:** SHAP reveals that the same feature can have different impacts for different students, supporting personalized intervention strategies.

#### 4.1.10 Limitations and Future Directions

- SHAP explanations are most reliable for tree-based models; deep neural networks may require alternative explainers.
- Further work could explore local explanations for individual at-risk students, or compare SHAP results across different model families and stages.

#### 4.1.11 Conclusion

SHAP analysis provides a transparent, quantitative framework for understanding model predictions. By surfacing the most important risk factors, it enables data-driven, interpretable, and targeted interventions to reduce student dropout.

## References

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